



ΣΧΟΛΗ ΨΗΦΙΑΚΗΣ ΤΕΧΝΟΛΟΓΙΑΣ ΤΜΗΜΑ ΠΛΗΡΟΦΟΡΙΚΗΣ ΚΑΙ
ΤΗΛΕΜΑΤΙΚΗΣ

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ΓΙΑ ΤΟ ΜΑΘΗΜΑ

"Μηχανική Μάθηση και Εφαρμογές"

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[Μέρος 1: Χρήση του GPT-2](#)

1: Παραδείγματα παραγωγής κειμένου του μοντέλου gpt-2

```
Prompt: > I enjoy walking with my cute dog  
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.  
I enjoy walking with my cute dog!!! I am so excited! Thank you so much and I really hope you will do a good job. Thank you and stay away from this post!
```

```
Prompt: > Hello
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Hello?"

"No, the man in red had no clothes. He was wearing an expensive suit and was wearing a white hat."

It was also revealed that, even though the man was actually the one who came from the city of Mogensha...he would probably have been killed by a demon if it were

The man's body was extremely thin, and that of the royal family was considerably thinner than the man's.

And the man who was the one who came from Mogensha was probably the man who would be the most wanted in the city...

"You're wrong, it's because you have something special that you've never really seen."

"What's that?"

"If I was the king and I saw what you were doing, I would have thought you were going to become King Duan. It's because of the fact that you brought this power
```

Παρατηρούμε πώς δίνοντας σαν prompt “ I enjoy walking with my cute dog”, “Hello“ ,το μοντέλο παράγει κείμενο, αλλά το περιεχόμενο είναι ασαφές.

Ακόμα και όταν εισάγουμε κείμενο από κώδικα το μοντέλο αδυνατεί να προβλέψει μία σωστή απάντηση

```
Prompt: > for i in range 1 to 11
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
for i in range 1 to 11)

0.5 m:1

0.5 m:2

0.5 m:3

For all other measurements, see the page about "How Do I Choose a Diameter of 4mm"?

Measurement for the 2.5mm diameter is as follows :
```

2: Διάφορες στρατηγικές αποκωδικοποίησης

Greedy

Χρησιμοποιώντας την τεχνική greedy επιλέγεται η επόμενη λέξη με βάση την μεγαλύτερη πιθανότητα. Στα παρακάτω παραδείγματα παρατηρούμε επαναλαμβανόμενο κείμενο.

```
Prompt: > I enjoy walking with my cute dog  
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.  
I enjoy walking with my cute dog and I love to play with my dog. I love to play with my dog and I love to play with my dog.
```

```
Prompt: > Hello  
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.  
Hello!"
```

```
"I'm sorry, but I'm not sure if you're ready to go."
```

```
"I'm not sure if you're ready to go."
```

```
"I'm not sure if you're ready to go."
```

```
"I'm not sure if you're ready to go."
```

```
"I'm not sure if you're ready to go."
```

```
"I'm not sure if you're ready to go."
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"I'm not sure if you're ready to go."
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"I'm not sure if you're ready to go."
```

```
"I'm not sure if you're ready to go."
```

```
"I'm not sure if you're ready to go."
```

Beam Search

Χρησιμοποιώντας beam search το μοντέλο κρατάει κάποιες διαδρομές (αυτό το δείχνει η παράμετρος num beams) και επιλέγει την διαδρομή που μεγιστοποιεί το γινόμενο των πιθανοτήτων των λέξεων της. Δεν εγγυάται όμως ότι θα βρει την καλύτερη διαδρομή πάντα. Παρατηρούμε πάλι το κείμενο να επαναλαμβάνεται.

```
➤ Prompt: > I enjoy walking with my cute dog
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
I enjoy walking with my cute dog!!!
```

```
I love walking with my cute dog!!!
```

```
I love walking with my cute dog!!!
```

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I love walking with my cute dog!!!
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I love walking with my cute dog!!!
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I love walking with my cute dog!!!
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I love walking with my cute dog!!!
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I love walking with my cute dog!!!
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I love walking with my cute dog!!!
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I love walking with my cute dog!!!
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```
I love walking with my cute dog!!!
```

```
I love walking with my cute dog!!!
```

```
Prompt: > Hello
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Hello!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
```

```
Prompt: > How are you ?
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
How are you?
```

```
What are you?
```

```
What are you?
```

```
What are you?
```

```
What are you?
```

```
What are you?
```

```
What are you?
```

Top-k search

Ο αλγόριθμος top-k επιλέγει τυχαία από τις top-k λέξεις την επόμενη με βάση την κατανομή των πιθανοτήτων. Στο παρακάτω παράδειγμα ο αλγόριθμος κρατάει τις top 50 λέξεις και από αυτές επιλέγει τυχαία την επόμενη.

```
Prompt: > I enjoy walking with my cute dog
Setting 'pad_token_id' to 'eos_token_id':50256 for open-end generation.
I enjoy walking with my cute dog ikis and catching my dog, but I really really really like to have a chat with this beautiful and gentle animal. In fact, I would rather have me
Anyway, at first I did a bit of a mental picture swap, and I ended up with the perfect image for my dog. The two pictures below were taken between the two of my dogs, and we be
This puppy is loving here at home with its mother, loves going swimming with cute children and is loving to go for walks. It isn't just a dog but a dog owner who helps her out
```

```
Prompt: > Hello
Setting 'pad_token_id' to 'eos_token_id':50256 for open-end generation.
Hello _____, and the list of things that I just wrote should include you all.
```

```
Prompt: > How are you ?
Setting 'pad_token_id' to 'eos_token_id':50256 for open-end generation.
How are you? Let's say I am a programmer.

Let's assume I have an application and I have a web service that offers service providers web services using HTTP/2. Then I would be running this program.

Let's build this app (assuming the code you build is ready for use!).

I'm going to start from scratch with just one step. I hope you get used to the tools that I offer to help you take advantage of those tools.

If I can figure out an easy tool, I'll put it in the next chapter. I encourage you to check out my other tools in the "Learn How" section.

After this, the project is now started by clicking the "Build and run a program with no dependencies" button that I created in the previous step.

There's another way to do it, which is to download the source code you just created. I'm going to link this link to the GitHub repository as well as put it
```

Παρατηρούμε τώρα ότι έχουμε ένα κείμενο που μοιάζει πολύ κοντά σαν να γράφτηκε από άνθρωπο. Παρόλα αυτά το κείμενο αυτό δεν έχει πολύ συνοχή. Εισάγωντας όμως ένα βαθμό τυχαιότητας παίρνουμε μέχρι τώρα τα καλύτερα αποτελέσματα.

Top-p

Ο top-p αλγόριθμος αναζητά την καλύτερη επόμενη λέξη τυχαία από ένα (ταξινομημένο με βάση την πιθανότητα) σύνολο λέξεων που έχει αθροιστική πιθανότητα μεγαλύτερη του p. Στην παρακάτω περίπτωση αναζητούμε σε κάθε βήμα την καλύτερη επόμενη λέξη που καλύπτει σαν άθροισμα πιθανοτήτων το 95%. Εδώ δεν έχουμε σταθερό αριθμό λέξεων όπως στην περίπτωση του top-k, αλλά το σύνολο των πιθανών επόμενων λέξεων διαφέρει ανάλογα με την πιθανότητα.

```
I enjoy walking with my cute dog (Umm.. yeah there are different owners
but I'm all about what I do and how I take care of my kids 'cause I love
them too)Some of you might not know, starting off with parenting is pretty
hard, especially with children. They're stubborn, big and cuddly, and with
them it can be overwhelming and I have to deal with that. Here I am taking
care of my toddler daughter with my deft hand and I'm hoping to ensure that
this experience has been accomplished. I'm also thrilled to say that
Cotswolds comes so well to knitting and knitting, I'd like to bring
Cotswolds into the 3rd cycle of my 2nd row of cycles, then continue to be a
regular contributor. Το κείμενο που παράγεται πάλι φαίνεται να είναι πιο κοντά στην
πραγματικότητα, αλλά χωρίς συνοχή.
```

Μέρος 2: Προσαρμογή μοντέλου

1. Παραδείγματα διαλόγων από την εκπαίδευση του μοντέλου

```
0: . Comeuing.  
  
!No.  
'm..  
? we have.  
  
't I... is? I  
.  
't he in what. You you be  
contin him me of IWhat I!? I't this this  
  
Wan, We I've! have'll. a I're a be in(  
  
II! don...'t I'll a.I I I it of you're will. you't'm are! you. I'm this.  
I(  
you.'m...Yesuing(, I'm IHe's'm you  
?!contin?I it I.He  
I?
```

```
0: ids,He't me.  
I'll. I'm do.  
We is is it.  
No to I don't not know. We're are get.  
There't the worry. Get!  
Let to get be do don't want'll going.  
You want. I think is get can't he know. He'll't your. Now.  
I're't? Your't the worry,
```

Στα παραπάνω παραδείγματα διαλόγων φαίνεται η υπερεκπαίδευση του μοντέλου καθώς δεν υπάρχει συνοχή κειμένου.

2. Δοκιμή διαφορετικών prompts στο εκπαιδευμένο μοντέλο

Στο παρακάτω παράδειγμα δοκιμάζουμε σαν prompt: “may the force be with you” και παίρνουμε 3 απαντήσεις response_text_0, response_text_1, response_text_2, οι οποίες φαίνεται να είναι άστοχες με βάση το περιεχόμενο των star wars. και τελείως διαφορετικές μεταξύ τους.

Prompt: > may the force be with you

Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.

response_text_0: may the force be with you?"

"Ma'am!" he said.

He fled himself up the path of a pile of stones in the midst of a thicket of sand and mightier dirt. He had not run a single foot, except by riding on himself and his Girdle. When the eighteenth morning light turned on, he did not return. The house opened a few steps from the low bridge to the next stream, and increased the depth of the river by five hundred feet.

To hold the highest care of men, he used to lure them with ropes or as if they were mighty crows. With one thrown on by another, he bargained quietly. One man found his prey through a ruined window in the wine cellar, by waiting there as he had been used to never seeing a man alone. He laughed as he saw a tall white stone with very soft yellow accents creeping up the waterfall. Another simply plucked the stone, which he recognized as a man's friend.

response_text_1: may the force be with you?" Which given my hour already, I have told and agreed to the usual treasonous ifstrytable raffing. I was then in my train o'.n the theatre, when the old station called into view, and upon it took us for some half an hour to get us out at Long Walter's Quay. I was then seen in my car, by his cousin Tavy Vott, to-night, when to his great surprise I saw him somehow up another hill up on the Q and made off. I saw through my neighborhood Hall's wall the place we had stumbled upon – probably, after our journey. There took us Ierway, though, that evening on my way up the looking-board, for so down we meet the Cape,* and then the twentys-five-quarters of the evening ; when we lay down on clits, frozen up and stale, and so out, they said, I shall, if we stay the whole hour, take

response_text_2: may the force be with you?"

21. "I'll see you next week."

*

02.41.9913 75 7:06 am.

MUMBAI RAHILTA: "I'm already going."

Mumana Sailhash to "Updating and Testing!" Everyone undressed

02.43.0013 74 7:06 am.

MUMBAI RAHILTA: "Hey there."

Sarasaka Rubikhi to "Encycling 1–"

02.46.0013 73 7:06 am.

Sarasaka Saukki Sukhin Heavycycle. Subject Satya was walking across the trail

Sarasaka Dalasa to "Taking Your Sunlight Blood and a bottle of Sunlight for a spin."
Satya threw drinks

02.47.0013 72 7:06 am.

Στην συνέχεια δοκιμάζουμε το prompt “for the republic” και παίρνουμε 3 απαντήσεις response_text_0, response_text_1, responsxe_text_2. Πάλι παρατηρούμε πολύ άστοχα αποτελέσματα σε σχέση με το περιεχόμενο των star wars.

Prompt: > For the republic

Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.

response_text_0: For the republic of Cuba, it is an appalling pattern of injustice that has been perpetuated. The New York Times, The Guardian, The Free Press, and the New Republic, almost all of them with their coverage and life, pretends that Cuba exists as human rights state or the rather terrifying concept of 'Cuban tyranny'.

Belatedly, this failing has been exacerbated by the fact that the European Union and the East Asian countries have committed to achieve 'Comandante Compliance Agreement', one of the extremely strict and repressive laws of the 20th Century, to which their country has been sworn.

It is worth emphasizing that the US and the European Union imposed their 'conversion' (Muslim servitude) order on Cuba as recently as 2011. The USA Commission recognized at the start of September 2011 that Cuba is not part of this Eurocentric order so the EU imposed import protection laws as its own draconian version of historical violations.

Such a horrific defense of

response_text_1: For the republic opponents say that the restriction has been imposed on themselves by others because they were marginalized in society in front of their father or grandfathers. Ironically all of the men facing these charges, including John Stones from Minnesota: the veteran defence lawyer, for example (who helped Perrin win the case) said he stood firm, claiming that he did not see any way that Perrin, as a criminal defendant, could play any role in the family morality. He encouraged his colleagues to say they were supporting out of duty and so it must not have been disingenuous to them. Perrin's use of the terms 'not allowed' and 'involvement' in the family court. The extent to which this ignores the significance of Perrin's involvement in these bylaws was key. The protections against torture and punishment, agreed with by Washington State, fell under Washington Sec. Clinton's purview. They ended with the peremptory trilemma of closing

response_text_2: For the republic National Geographic put their illustration out to their fans (oh not too big on good stuff) on their site, that succinct and honest image was so widely shared on board, that after a Goofy "Let's Take a Big Person Out to the Books" guy praised their "useful sketches," he was told to check it out first, to make sure no one was playing with his work. As New York Magazine's cover variant explained: "He's tall and has a singular, but hilarious sense of humor. Straightening stubble and length are essential. Anyone can do the job here."

Δοκιμάζοντας σαν prompt: "I'm sure Luke wasn't on that thing when it blew. He wasn't. I can feel it." (ένα κομμάτι διαλόγου) , παρατηρούμε ότι σε κάποια σημεία έχουμε σχετικά κοντά αποτελέσματα με βάση το πλαίσιο των star wars(response_text_1) , αλλά γενικότερα πάλι έχουμε αστοχίες.

Prompt: > I'm sure Luke wasn't on that thing when it blew. He wasn't. I can feel it.

Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.

response_text_0: I'm sure Luke wasn't on that thing when it blew. He wasn't. I can feel it. Actually, this is pretty silly. I said back then that most pro athletes in the NFL would step down now. Do you know any athletes now who were still with that organization? And can you get through that? OK, as I said back then, I just have to figure out for myself. Here's the thing, I love the way everybody is handling this, just kind of dealing with that stuff and going with the flow, taking a step back in time when you hear what Kevin Durant's saying in that book. You know, the thing that's getting in front of us today is he's gonna say something and that's good.

On getting into better relationship with the Bill Simmons organization on America's Most Wanted

JG: I was wondering what was the plan to do that when you were talking about us years ago and saying we wouldn't go out and do anything else that we thought would work. Couldn't have done it? Couldn't have done it?

response_text_1: I'm sure Luke wasn't on that thing when it blew. He wasn't. I can feel it. God knows what else happened to him. Let's call it the accident."

Yeah, and there's where this incredible moment occurred. As Luke was frantically removing weapons from the interrogation and beginning his rounds, he suddenly forgot himself; it's not like that guy actually hits Luke so hard or anything. Chris had started rushing out of the way just out of a three-way stop, which meant that he wasn't even at all aware that Luke had been killed.

In a voice from the chamber, Luke said, "Shit! To what?!"

The Bounty Hunter looked back at Chris with awe. "Ahh, hi, Cody. I was the lookout for a suspicious thing inside the intergalactic mining ring. I should've been an extra. Right, let's get down in this lab next to the stasis chamber, and see if that's some sort of lightsaber."

"I knew it," Cody said, "But why was it there?"

response_text_2: I'm sure Luke wasn't on that thing when it blew. He wasn't. I can feel it. He never said it, did it. It didn't take me five seconds to put into words the words he lived by, "Oh wow." He would connect things. He would connect things; he'd connect the things that worked in his head like jigs, nuts, bolts, bolts, bolts, bolts. And that's all stuff I knew, but didn't know the hard way. His strength and that had made him that great person that I was, way beyond his capabilities in any way. That was always an inside joke; things worked out really nicely on his plate so it was something I did every day. He had a sweet cap out there, and if he didn't catch his breath when I mentioned he couldn't fly a motorcycle into space or make it past certain corners, I would add that I have faith it would work for him.

3.Ερμηνεία παραμέτρων μοντέλου GPT-2

```
➡ The GPT-2 model has 148 different named parameters.

==== Embedding Layer ====

transformer.wte.weight          (50257, 768)
transformer.wpe.weight          (1024, 768)

==== First Transformer ====

transformer.h.0.ln_1.weight      (768,)
transformer.h.0.ln_1.bias        (768,)
transformer.h.0.attn.c_attn.weight (768, 2304)
transformer.h.0.attn.c_attn.bias (2304,)
transformer.h.0.attn.c_proj.weight (768, 768)
transformer.h.0.attn.c_proj.bias (768,)
transformer.h.0.ln_2.weight      (768,)
transformer.h.0.ln_2.bias        (768,)
transformer.h.0.mlp.c_fc.weight  (768, 3072)
transformer.h.0.mlp.c_fc.bias    (3072,)
transformer.h.0.mlp.c_proj.weight (3072, 768)
transformer.h.0.mlp.c_proj.bias  (768,)

==== Output Layer ====

transformer.ln_f.weight          (768,)
transformer.ln_f.bias            (768,)
```

Embedding Layer

transformer.wte.weight: 50259 δηλώνει το μέγεθος του embedding. Κάθε λέξη αναπαρίσταιται με ένα διάνυσμα [768:1]

transformer.wpe.weight: Αναφέρεται στο positional encoding. Το 1024 δηλώνει το μέγιστο input που μπορεί να πάρει το μοντέλο * 768 που είναι το μέγεθος κάθε διανύσματος

First transformer

transformer.h.0.ln_1 weight και transformer.h.0.ln_1.bias: Αναπαριστούν τα βάρη και τα biases για το normalization μετά το self-attention mechanism.

transformer.h.0.attn.c_attn.weight και transformer.h.0.attn.c_attn.bias: Αναπαριστούν τα βάρη και το bias για τους γραμμικούς μετασχηματισμούς που εφαρμόζονται στις εισόδους των embeddings στο self-attention layer (3*768->2304 το οποίο αναφέρεται στα διανύσματα K,Q,V)

transformer.h.0.attn.c_proj.weight και transformer.h.0.attn.c_proj.bias: Αναπαριστούν τα βάρη και τα biases για τον γραμμικό μετασχηματισμό των εισόδων του αποκωδικοποιητή μετά τον μηχανισμό προσοχής.

transformer.h.0.ln_2.weight και transformer.h.0.ln_2.bias: Αναπαριστούν τα βάρη και τα bias για το normalization μετά την έξοδο του νευρωνικού δικτύου.

transformer.h.0.mlp.c_fc.weight και transformer.h.0.mlp.c_fc.bias: Αναπαριστούν τα βάρη και τα biases του 1ου επιπέδου στο νευρωνικό δίκτυο

transformer.h.0.mlp.c_proj.weight και transformer.h.0.mlp.c_proj.bias: Αναπαριστούν τα βάρη και biases του 2ου επιπέδου στο νευρωνικό δίκτυο

Output layer

transformer.ln_f.weight και transformer.ln_f.bias: Αναπαριστούν τα βάρη και τα biases για το linear επίπεδο, πριν την συνάρτηση softmax.